

Assignment

1. What is operating system? Explain the types of operating system.
2. Describe the functions of operating system.
3. What is kernel? Describe the types of kernel.
4. Explain virtual machine and client server model in operating system.
5. What is process? How it is differ from program? Explain.
6. What are different process models? Draw a state (block) diagram of process with different state and explain each briefly.
7. What is multithreading? Explain.
8. Compare the use of monitor and semaphore operations.
9. Explain dining philosopher problem.
10. For the process listed in following table, draw a Gantt chart illustrating their execution using:
 - a. First-Come-First-Serve
 - b. Short-Job-First
 - c. Shortest-Remaining-Time Next
 - d. Round-Robin (quantum = 2)

Process	Arrival Time	Burst Time
A	0.00	4
B	2.01	7
C	3.01	2
D	3.02	1

11. What is deadlock? State the conditions necessary for deadlock to exist. How can you represent a deadlock? Explain with suitable example.
12. Describe various methods for deadlock prevention?
13. Explain Resource Allocation Graph in detail.
14. Assume that there are three resources, A, B, and C. There are 4 processes P₀ to P₃. At T₀ we have the following snapshot of the system:

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₀	1	0	1	2	1	1	2	1	1
P ₁	2	1	2	5	4	4			
P ₂	3	0	0	3	1	1			
P ₃	1	0	1	1	1	1			

- a. What is the content of the matrix need and Total Resource Vector?
 - b. Is the system in a safe state?
15. Given three process A, B and C and three resources X, Y, and Z are following events happen in the sequence as given:

A request X

B request Y

C request Z

A request Y

B request Z

C request X

C request Y

Assume that requested resource should be always be allocated to request process if it is available. Draw the resource graph for the sequence and also mention whether it is deadlock? If it is how to recover the deadlock?

16. Explain the term logical address and physical address.
17. Describe fragmentation with example.
18. What is paging? Describe types of page tables.
19. What is segmentation? How can it be used for the memory management?
20. What is TLB? How can it be used in paging? Explain.
21. What are different page replacement algorithms? Explain
22. Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6.
How many page faults would occur for the FIFO, LRU and optimal page replacement algorithm? Assuming four frames, remember all frames are initially empty.
23. Given five memory partitions of 100KB, 500KB, 200KB, 300KB and 600KB (in order), how would each of the first-fit, best-fit and worst-fit algorithms place processes of 212KB, 417KB, 112KB and 426KB (in order)? Which algorithm makes the most efficient use of memory?
24. Describe the term programmed driven I/O and Interrupt driven I/O.
25. What is DMA? Write the working procedure of DMA.
26. Define the term seek time and rotational delay in disk scheduling. Suppose that a disk has 100 cylinders, numbered 0 to 99. The drive is currently serving a request at cylinder 43 and previous request was at cylinder 25. The queue of pending request, in FIFO order is: 86, 70, 13, 74, 48, 9, 22, 50, 30.
Starting from the current head position, what is total distance (in cylinders) that the disk arm moves to satisfy all pending request for each of following disk scheduling algorithm.
a. FCFS b. SSTF c. SCAN d. LOOK
27. What is file? Write down their importance in operating system.
28. What is file types? Describe file types used in operating system.
29. Explain different structure of directories.
30. Why access protection is necessary in file systems? How protection can be implemented by the identity of the users?
31. What do you mean by user Authentication? Explain various user authentication methods with example.
32. How can we secure our system? Explain.
33. What is distributed operating system? Describe its features.

34. Describe types of clock synchronizations with examples.
35. Perform CASE Study on windows and Linux operating system (must include history, features, advantages and disadvantages of each operating system).